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Work Package 6

Early estimates of economic indicators

Deliverable 6.7

Report and recommendations about the methodology and process of calculating estimates for at least one early economic indicator

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1 Summary

The main achievement of WP6 was to show that how a combination of (early available) multiple big data sources, administrative and existing official statistical data could be used in creating an existing or new early estimates for official statistics.

Here we provide concrete advice on how to embrace the opportunities of nowcasting. We show, that NSIs can shorten the publication lag in a straightforward way without compromising. We offer a range of methodological recommendation related to methodology being investigates for purposes of nowcast of early economic indicators using traffic loop data.

During the ESSNet project in the SGA-1 period some nowcasting methods for the purpose of estimating Turnover Indices were discovered and tested. The original purpose was to explore the possibilities of estimating the consumer confidence index and (or) indices of turnover. Due to the early findings about the inaccessibility of social media data, which is crucial for assessing the consumer confidence index, the WP6 team focused on Turnover Indices.

This study contains the report and recommendations about the methodology and the process of calculating estimates discovered estimating Turnover Indices at Statistics Finland and the Statistical Office of the Republic of Slovenia (SURS).

With the help of Statistics Finland SURS prepared the application (together with instructions how to use it) which allows inputting different kinds of data, testing various conditions for choosing principle components, producing quality indicators which compare results of different nowcasting methods and producing quality indicators which compare results with disseminated official statistics. This study describes the methodology and process of nowcasting indicators from the point of data acquisition to the end results on GDP and also on a known GDP correlator, the Industry Turnover Index. Imputations of missing count spots data received a big consideration in this work.

Statistics Finland used more than one hundred models in order to compute early estimates of the indicators of interest. In this study they discuss the generalities of the model classes explored, and then provide a list of the models which were incorporated in the forecast combinations, an approach which provides the best performance and is therefore the preferred methodological choice in nowcasting with the type of data employed. Some practical considerations related to the overall methodological framework are discussed. All the models used are easily implementable, in terms of software used and coding efforts. This study also evaluates the quality in terms of timeliness and accuracy in capturing the target indicators' next values.

In the Netherlands, a different approach was developed. Due to the large number of traffic loops (20,000 on the Dutch highways), another data cleaning strategy had to be developed. To enhance the speed of the data cleaning process, a Monte Carlo Markov Chain (MCMC) algorithm was used to clean the data, remove noise and solve the problem of missing data. The neural network is able to deal with missing and erroneous values, which makes it ideal for data cleaning and dimension reduction at the same time. By training subsets of road sensors in different neural networks, the dimensionality problem as described above is solved. The reduced dataset can then be used for regression purposes.

NSIs can address a major quality issue, namely the timeliness, by using a range of micro level data sources accumulated in the registers well before the official release is made, by employing large dimensional econometric models, to form an initial quick estimate of the target indicator.

We have shown that this does not necessarily lead to too large revisions, but adds significantly to the quality of official statistics through timeliness dimension.

This line of work can proceed in multiple directions:

- Other data sources can be explored with the methodologies we have presented and possibly in relation to other indicators
- Other modelling frameworks are possible
- A real-time application can be programmed, especially relying on the traffic loops data
- These methodologies can be implemented into the production systems of official statistics, and their added value is not limited to nowcasting the GDP or some aggregate final indicators, but could be explored in order to impute some missing components of the aggregated figures.

Such approaches can be easily implemented across the entire European Statistical System.

2 Nowcasting economic indicators using Principal Component Analysis method at SURS

Nowcasting is a very early estimate produced for an economic variable of interest over the most recent reference period calculated on the basis of incomplete data using a statistical or econometric model different from the one used for regular estimates. Soft data should not play a predominant role in nowcasting models. Nowcasts may be produced during the very same reference period for which the data are produced.

Conducting one of the pilots during the SGA-1 ESSnet project several nowcasting methods were under investigation. For the first stage (dimensionality reduction), the most promising in the sense of practical implementation was the Principal Component Analysis method. The central idea of principal component analysis (PCA) is to reduce the dimensionality of a data set consisting of a large number of interrelated variables, while retaining as much as possible of the variation present in the data set. This is achieved by transforming to a new set of variables, the principal components (PCs), which are uncorrelated, and which are ordered so that the first few retain most of the variation present in all of the original variables. For the second stage (statistical learning), many methods were tested (e.g. linear regression). The idea of statistical learning is to learn from a training set of data and infer the function which is validated on the test set of data and can be used to predict future output from future input.

The main idea is to practically use at least one of the nowcast methods for the purposes of estimating early economic indicators.

The PCA Model [i] consists of two stages:

1. Principal component analysis (PCA)

- PCA is used for dimensionality reduction of microdata (e.g. enterprise data)
- First, time series of microdata are standardized, then, first few principal components are chosen
- Various conditions for choosing principal components are tested:

  - The chosen principal components explain at least 70% (75%, 80%, 85%, 90%) of variability of enterprise data.
  - Time series in the linear regression model are at least 7 (8, 10, 15, 20) times longer than the number of the chosen principal components.
  - The last chosen principal component explains at least 5% of variability of microdata.

2. Linear regression(LR)

- Linear regression is used for statistical learning. Regression coefficients, estimated on data until time T-1, are used to estimate the dependent variable at time T. Regressors are standardized.
- The dependent variable Y is our time series of interest (e.g. turnover index).
- The regressors $X_1, X_2, \cdots, X_k$ are the chosen principal components; some other regressors can be added (e.g. aggregated traffic sensor' data).

SURS with the help of Statistics Finland prepared application (together with instructions how to use it) which allows:

- inputting different kind of data
- testing various conditions for choosing principle components
- producing quality indicators which can be used to compare results of different conditions
- producing quality indicators which can be used to compare results with disseminated official statistics

At the Statistical Office of the Republic of Slovenia we calculate the statistics of gross domestic product (GDP) every quarter of a year. 670 days after the reference period is unfortunately the quickest we can publish such statistics as the timeliness of GDP data is limited by the survey evaluations of some of the components that make up GDP. However the use of flash (rapid) estimates could fasten this process. On the basis of investigation of various big data sources we had the idea to use the data we acquired from traffic sensors and use them as primary regressors (PCA is used for traffic sensors' microdata and first few principal components are chosen for regressors) andor secondary regressor (aggregated traffic sensors' data are used as regressors) in a linear regression method for nowcasting GDP 45 days after the end of the reference period. Nowcasting is a method of calculating estimates on the basis of unknown present or near-future values with the use of a known correlator. Further on we describe the methodology and process of nowcasting indicators from the point of data acquisition to the end results on GDP and also on a known GDP correlator, the Industry Turnover Index.

2.1 Data preparation and editing

Data were acquired from the Ministry of Infrastructure in bulk for now. The first dataset we received was yearly tables of traffic through 647 sensor spots on Slovene regional and speedway roads. The counting intervals are 15 minutes long. The dataset included data from the start of 2011 to the end of 2016. This dataset was acquired in May 2017. The second dataset was received in February 2018 for the data of 2017 in the same format as above. In the future we have plans to move towards a timelier transfer process, but for the purposes of analysing and using the data in an experimental statistic, we deemed these transfers enough. The 15-minute intervals are the shortest intervals available at the Ministry.

The data we acquired were partly edited already: the actual files contained aggregated numbers of multiple sensors on the same stretch of road if they existed. We named these objects count spots.

2.2 Data Imputation

Imputations had received a big consideration in our work. We focused on regional roads and the imputations are tailored towards them. The imputations were used when missing data were

detected and only after aggregation to a monthly basis was done. This was done in order to avoid precision errors due to small scale data (one to two cars more in one period may represent growth of 100% in the mornings, while this may only be 1-2% growth in the peak hours) and to limit the data size on disk to a manageable degree. Our imputations of a count spot data are based on the growth of neighbouring count spots.

The possible errors that we detected or anticipated were errors that happen due to algorithmic errors and errors that occur due to the differing scales of traffic in count spots in comparison to their neighbours' traffic. An extensive analysis of scale and shape of data and performance tests of 4 different algorithms were concluded.

We tested various methods of imputation. The chosen algorithm was tested on a subset of data with non-missing values, and turned out to be quite accurate. On regional roads the error was less than 5% on an imputation of a full month, while the error on speedway sensor data was a bit higher. But, as mentioned above, the speedway data was not as important to us, as we supposed they contain information impertinent to our measurements.

2.2.1 Developing of imputation method

Edited data were imputed only in case of traffic sensor failure, depending on the traffic sensor location: speedway, highway or a regional road. From the entire population of count spots we removed the count spots on motorway connections.

The imputation was performed on monthly aggregated data. In case when less than 20 days of aggregated data were missing, only these data were imputed relative to growth over the same period; otherwise, complete monthly aggregated data were imputed. The following notation was used:

S is the set of all count spots;

$g_k^{m,t}$- is the number of ***all observable*** traffic of one vehicle category through count spot k in the month m of year t – this amount is often unknown to some degree, due to failure or inactivity of the sensors in the count spot, and imputations try to estimate it;

$d_k^{m,t}$ is the share of periods with ***existing/observed*** data in the month m of the year k through count spot k, and this number is either 0 or $\frac{days\ in\ motnh-20}{days\ in\ month} \leq d_k^{m,t} \leq 1$. It stands to reason that months that need imputation have this share <1;

$imp\left(d_k^{m,t} g_k^{m,t}\right)$ is the imputed value;

s_k is a neighbour of the count stop k;

$s(k)$ is the set of neighbors for the count stop k, defined as $s(k) = \left\{ s_i;\ i = 1 \ldots 4 | s_i = \arg\min_{\substack{s_i \in S \\ s_i \neq k}} \left(\Delta_e(k, s_i)\right) \right\}$;

K' is a subset of the neighbours, which adheres to some conditions; and

$\Delta_e(k, s_i)$ is the Euclidian distance between count spots k and s_i.

The basic imputation idea was to use the traffic growth of the nearest count spot. In doing so, we avoided imputation from imputed data. When the nearest neighbour contained missing data in the same period, we searched for the next nearest neighbours and tried to impute data with them, and so on. This procedure was carried out until the fourth nearest count spots.

In the mathematical record, the neighbours were defined as the four nearest count spots with respect to the Euclidean distance between the geographic coordinates of the counting position and neighbour:

$$s(k) = \left\{ s_i;\ i = 1 \dots 4 | s_i = \arg\min_{\substack{s_i \in S \\ s_i \neq k}} \left(\Delta_e(k, s_i)\right),\ \Delta_e(k, s_i) = \left(x_k - x_{s_i}\right)^2 + \left(y_k - y_{s_i}\right)^2 \right\}$$

where x_l and y_l represents geographic coordinate of one count spot l.

If we did not find such a neighbour, we used the same procedure one more year before. If the suitable neighbour was still absent, the count spot was imputed manually. We requested the existence of non-imputed data in the selected year also for the current count spot. In the following formulas, this is expressed by a set of conditions $Z^d = \{d_{s_k}^{m,t} = 1,\ d_{s_k}^{m,t-l} = 1, d_k^{m,t-l} = 1\}$ where $t - l$ represent previous or previous year in relation to the year t.

The values of count spots in 2011 (first observed year) and some in 2012, when the count spot in 2011 was imputed, were imputed with respect to future years. The first reviewed way focused on the monthly growth of an individual neighbour.

2.2.1.1 Growth of an individual neighbour

First the data for the neighbour during the searched period are checked. If the data exist, the data of the same period of the previous year are checked. If in the previous year it happened that there were neither data in the adjacent count spot or any vehicle of a category, we cannot count growth for this category. In such cases, the basis for the imputation is the unchanged value of the previous year of the count spot, which is imputed. If, however, the neighbour's value is positive in the previous year, we calculate growth and multiply it with the imputation base according to the following formula

$$imp_1\left(d_k^{m,t} g_k^{m,t}\right) = \\ = \begin{cases} \left(\frac{g_{s_k}^{m,t}}{g_{s_k}^{m,t-l}} g_k^{m,t-l}\right)\left(1 - d_k^{m,t}\right) + d_k^{m,t} g_k^{m,t};\ g_{s_k}^{m,t-l} \neq 0, s_k = \arg \min\limits_{\substack{s_k \in s(k) \\ Z^d\ za\ s_k}} \left(\Delta_e(k, s_k)\right), \\ g_k^{m,t-l}\left(1 - d_k^{m,t}\right) + d_k^{m,t} g_k^{m,t}; \qquad d_k^{m,t-l} = 1, \\ g_k^{m,t}; \qquad otherwise. \end{cases}$$

2.2.1.2 Average growth of neighbours

The basis is previous value of the count spot to which one data item is imputed and multiplying it with the average of the neighbours' growth. The mathematical formula is similar to the previous one, with the difference that this time we count growth on several neighbours and not just the first closest ones that satisfy the conditions.

$$imp_2\left(d_k^{m,t} g_k^{m,t}\right) = \begin{cases} \left(\sum\limits_{\substack{s_k \in K' \subseteq s(k) \\ Z^d \forall s \in K' \\ g_{s_k}^{m,t-l} \neq 0 \forall s \in K'}} \frac{g_{s_k}^{m,t}}{g_{s_k}^{m,t-l}} g_k^{m,t-l}\right) \frac{1}{|K'|}\left(1 - d_k^{m,t}\right) + d_k^{m,t} g_k^{m,t}, |K'| > 0, \\ g_k^{m,t-l}\left(1 - d_k^{m,t-l}\right) + d_k^{m,t} g_k^{m,t}; \qquad d_k^{m,t-l} = 1, \\ d_k^{m,t} g_k^{m,t}; \qquad otherwise. \end{cases}$$

2.2.1.3 Total growth of neighbours

For the current count spot, we found up to four neighbours with non-imputed values in the current and previous periods, and then we summarized them by periods and divided these values. This relative difference represents better growth, more resistant to changes in traffic over periods.

$$imp_3\left(d_k^{m,t} g_k^{m,t}\right) = \begin{cases} \left(\frac{\sum_{\substack{s_k \in K' \subseteq s(k) \\ Z^d \forall s \in K'}} g_{s_k}^{m,t}}{\sum_{\substack{s_k \in K' \subseteq s(k) \\ Z^d \forall s \in K'}} g_{s_k}^{m,t-l}} g_k^{m,t-l}\right)\left(1 - d_k^{m,t}\right) + d_k^{m,t} g_k^{m,t}, \sum\limits_{\substack{s_k \in K' \subseteq s(k) \\ Z^d \forall s \in K'}} g_{s_k}^{m,t-l} \neq 0 \\ g_k^{m,t-l}\left(1 - d_k^{m,t}\right) + d_k^{m,t} g_k^{m,t}; \qquad d_k^{m,t-l} = 1, \\ d_k^{m,t} g_k^{m,t}; \qquad otherwise. \end{cases}$$

2.2.1.4 Imputation of count spots placed on speedways and highways

The basis is imputation of missing count spot data with values of the nearest neighbour. The requirements for finding neighbours are simplified in the sense that we find the appropriate neighbour in the counting direction. Since some of the counting directions are innumerable, we have found a neighbour who had the same information of the counting direction. This left us with cases where the imputation was not carried out over a period, because the count spot with both directions has only one-way neighbours.

2.2.2 Selection of imputation method

Imputation with the first method proved to be unreliable, as we often deal with less busy roads, where small absolute changes (for example, traffic in the previous period of 31 vehicles and in the next period 59 vehicles) quickly represent a large relative change. In addition, on such roads some

categories are so often present that imputed values are not calculated from the growth of the neighbouring count spot, but directly from the value of the previous period of the same count spot. For this reason, other imputations have been tested. The main idea was to take into account the data of several neighbours. With this, their possible high relative growth would have a lesser impact on total growth. When testing the influence of imputation (explained below), we found that imputation errors are small, and the standard deviation is around 9%.

The first idea, of course, was the average relative growth of all four neighbours. Unfortunately, this mode has worsened worse than expected, as there are many low-traffic neighbours with little traffic. This did not solve the problems of high relative growth, but we duplicated it at most.

The third way was to take the total traffic value of all neighbours and compare it with the values of the same count spot in the previous relevant period (the corresponding period is closest to the previous period, where all the data are present, at least for one neighbour and the considered count spot). With such imputations we solve the problem of large fluctuations growth, since the amount of traffic that we aggregate is higher in the share and divisor, and therefore more resistant to disruptions. In addition, we also reduce the number of neighbours we need to give up because they take the values 0 in the divisor. The mathematical basis for this calculation comes from the theory of graphs, since our thinking is equivalent to joining adjacent knots into a single knot. In order to take into account total traffic through several count spots, the standard deviation is lower than in the first mode (around 5%) and the average is of about the same quality (several times even a bit better). In testing the final result of imputation we reached the value which rarely deviated more than 5% in terms of real value. In certain categories, these deviations are, of course, greater (in both directions).

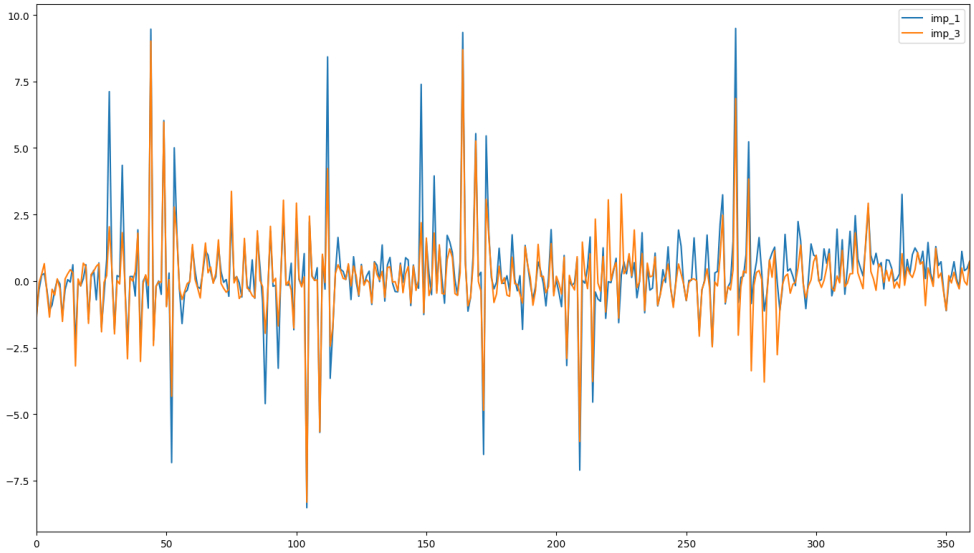
We implemented another method of testing. The data of one of the count spot were imputed in each month and then the relative difference in value between one 'imputed' count spot and the total original value of all five count spots was calculated.

With formula:

$$Eff_{l,K,i}^{m,t} = \left(\frac{imp_i \left(\widehat{d_l^{m,t} g_l^{m,t}} \right) + \sum_{\substack{k \in K \subset S \\ k \neq l}} g_k^{m,t}}{\sum_{k \in K \subset S} g_k^{m,t}} - 1 \right) \cdot 100\%,$$

Where $Eff_{l,K,i}^{m,t}$ is the influence of the value of the imputation of the count spot l in month m of year t with mode i to the total traffic value of all count spots in the subset of count spots K. The impact is expressed in percentages.

Picture 1: The relative difference (in %) is influenced by the imputation of one of the five count spots from the regional road in the monthly period



In Figure 6, we can see the actual imputation effects of an individual period of one count spot of the count at the total of 5 count spots, where two different methods were tested. The values for each count spot are given in each month and each year (in this order). In this case, we compare the count spots with the codes 054 (area of Trzin), 056 (area of Zreče), 061 (area of Maribor) and 062 (area of Ormož). The blue curve is the first way of imputation (imp_1), orange is the third way (imp_3). We can notice that the orange curve reaches lower extremes than the blue one, which means that the differences between the imputed and the original data are also smaller. In this case, the average value of the imputation effect of one period of a count spot on the values of all five counters is in this case $0.3016\%(\pm 1.8462\%)$ for the first method and $0.1152\%\ (\pm 1.5302\%)$ for the second method.

Figure 2: An Impact on a subset of five count spots (in%)

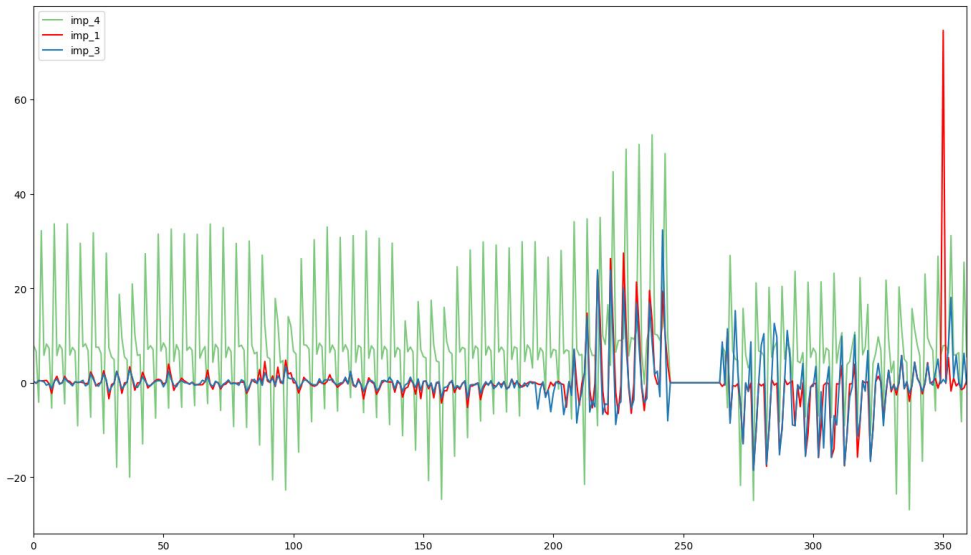
Python Console

```
>>> bla
     Leto Mesec Stevec                 %
0    2011     1    055    0.310178155355
1    2011     1    056   -0.472618659509
2    2011     1    057    0.278986055806
3    2011     1    060    -2.57421369832
4    2011     1    061    0.331194122592
5    2011     2    055     1.77822026952
6    2011     2    056    -1.29647915386
7    2011     2    057   -0.781748270187
8    2011     2    060   -0.268614672071
9    2011     2    061    -0.51575653205
10   2011     3    055     0.27589006258
11   2011     3    056    -0.47002713024
12   2011     3    057   -0.072408901537
13   2011     3    060    -1.70639314612
14   2011     3    061   0.0763927845157
15   2011     4    055     1.31771155166
16   2011     4    056  -0.0570475803762
17   2011     4    057   -0.370945187996
18   2011     4    060   -0.714871690195
19   2011     4    061   -0.261627414023
```

Figure 2 shows the impact of the imputation of each month on the total value of five count spots. In testing the influence of one category of imputed count spot with respect to the selected subset of all count spots (in practice, we used subsets of size 5 count spots) we found that the average imputation effect is rarely over 1% relative to the true value. However, these effects seem to be more positive than negative, so our method is not necessarily impartial. We decided to change this and choose the best of the presented ways: imputation with regard to the total growth of traffic neighbouring count spots.

According to the data of count spots on highways and motorways, the results of imputation tests are worse with all modes, which is why we tried an additional way of imputation, based on the idea of foreign drivers who are only crossing the country. Reflection is in the direction of maintaining the value of motorway values by category, as most vehicles enter the motorway at one border and exit at the other border. Maintaining the number by count spots allows us to directly insert the adjacent values multiplied by the share of the present data of the considered count spot. This idea is good, but in practice, count spots are too much away from each other in order to transport the traffic effectively. Due to the high number of motorway connections, the imputed values are often under- or overvalued. Due to the concentration of traffic around Ljubljana, imputed values are on average higher than the real values on all count spots that are not in this area but have neighbours there. On testing, absolute errors are often above 30% or even 100%. The standard deviation of errors proved to be equivalent to the third method in testing, but individual deviations of this mode are greater than the defects in the third mode, as can be seen from Figure 3.

Picture 2: Impacts of a single period imputation on a subset of meters on highways and motorways



We can see that imputed values with the fourth imputation method (green curve) are higher than the real ones (on average by 7.5211% of the true value), and the standard deviation is very high over the whole period (12.6224%). In comparison, the first and third methods are extremely precise, even in earlier periods. When comparing the average error and the standard deviation, the third imputation method (with -0.07210% is the average error and 5.3963% with standard deviations) is best shown. Nevertheless, the results of imputations at the count spots of motorways and speeds, due to larger deviations, are less accurate than the results of count spots on regional roads.

2.2.3 Analysis of imputation of missing data

We have chosen the third way of imputation imp_3 both for regional data as well as for data on highways and speedways.

The following table summarizes the overall results of imputation:

Table 1: Some general information on imputation in counters

	Count spots TOTAL	Count spots REGIONAL ROADS	Count Spots HIGHWAYS and SPEEDWAYS
Number of count spots used	440	398	42
Number of count spots used with missing data	333	295	38
Number of imputed periods (months)	1403	1178	225
Number of fully imputed periods	451	406	45

(months)			
Percentage of present 15-minute intervals	97.6175%	97.6997%	96.8383%
The amount of traffic before imputation	7786776222	5537534642	2249241580
The amount of traffic after imputation	8001973920.508	5662172401.306	2339801519.202
Change in traffic volume - absolute	215197698.508	124637759.306	90559939.202
Change in traffic volume - relative	1,028	1,023	1,040

The table shows that 2.8% of the value of traffic (2.3% on regional roads and 4% on motorways and express roads) was imputed, and on average, we added 2.25% of traffic in each year with imputations of count spots of regional roads and by 4.11% at count spots of highways and speedways.

2.3 Quality assessment of nowcasting

After the imputation was done, we used the final data as a secondary regressor in a nowcasting method. The method consisted of a linear regression model with principal component analysis (PCA) that was used to find the best fit of quarterly enterprise turnover data onto GDP values. Afterward, we predicted the next period with the optimal model.

We tested many combinations of regressors, by using a PCA model with parameters on the main regressors and adding traffic datasets as secondary regressors. We expected that the best result would be obtained when using cargo traffic on regional roads. Since Slovenia is a transit country, the so-called crossroads between East and West and North and South of Europe, our view was that the inclusion of highway traffic would be detrimental to our results. In order to justify our assumptions we used a dataset of full traffic data, a dataset of only cargo traffic data, a dataset of traffic on regional roads and a dataset of only cargo traffic on regional roads.

We then used these linear regression models to nowcast the estimates of GDP in three points in time (namely second, third and fourth quarters of 2106). The results are quite clear; the traffic data improve the quality of estimates in every occasion compared to solely using industry data , while the consistently the best PCA model seemed to be the model *80%*[1], which only uses eigenvectors of the main regressors with the highest eigenvalues that explain 80% of variability of data.

[1] The PCA method is used because the number of variables in microdata is too large for linear regression and therefore needs to be reduced. In our models the names *75%, 80%, 85%* and *90%* mean PCA conditions that take just enough first few principal components (variables) to explain 75%, 80%, 85% or 90% of the variability of the microdata. The chosen principal components are eigenvectors with the highest eigenvalues of the covariance matrix of the microdata.
The PCA method with PCA condition *po10* takes just enough first few principal components to have at least 10 times more time points from the beginning to the ending of the time span.
The PCA method with PCA condition *zadnja5* takes all eigenvectors (principal components) whose eigenvalues have at least a 5% share among all of the eigenvalues.

Table 1: Absolute maximum errors for different traffic count spots datasets

PCA method	Without traffic data (in million EUR)	All vehicle categories, all roads (in million EUR)	Cargo vehicle categories, all roads (in million EUR)	All vehicle categories, regional roads (in million EUR)	Cargo vehicle categories, regional roads (in million EUR)
75%	129.04	151..68	136.85	194.45	101.33
80%	149.46	154.75	146.71	105.55	42.58
85%	200.13	228.54	207.52	207.62	167.30
90%	131.46	232.92	136.62	204.39	106.43
po10	282.41	313.32	248.34	216.84	207.59
zadnja5	102.70	110.33	113.10	118.91	56.78

Source 1: Own

Table 2: Absolute mean errors for different traffic count spots datasets

PCA method	Without traffic data (in million EUR)	All vehicle categories, all roads (in million EUR)	Cargo vehicle categories, all roads (in million EUR)	All vehicle categories, regional roads (in million EUR)	Cargo vehicle categories, regional roads (in million EUR)
75%	108.44	120.71	111.17	111.67	83.94
80%	85.47	88.94	88.26	68.65	24.14
85%	115.30	134.35	119.59	150.90	111.14
90%	49.38	82.96	51.11	143.69	68.356
po10	255.25	225.39	211.97	144.45	174.05
zadnja5	66.96	72.78	66.87	48.17	37.20

Source 2: Own

As can be seen from Table 1 and Table 2, our assumptions about traffic data were right. In almost every instance any use of traffic data corrects the estimates a little. However, we were pleasantly surprised when we checked for the best combination. Use of any model with the exclusion of *90%* resulted in better accuracy when using cargo data on regional roads. The errors between the official values of GDP and our estimates are on average reduced by a factor of 4, compared to the non-traffic example. Furthermore, the maximum absolute error was around 2.5 times smaller when using traffic data in comparison to when not using traffic data. Even the best model without usage of traffic data was still worse than some of the models with included traffic data (but it must be said that we suspect that there was a high chance of overfitting in the best non-traffic model). Lower errors when comparing all roads to regional roads or all vehicle categories to cargo vehicles also support our assumptions.

As can be seen from Table 3 below, the relative errors of estimations when using traffic data are seldom more than one percent.

Table 3: Estimates and errors of no-traffic data and traffic data sets

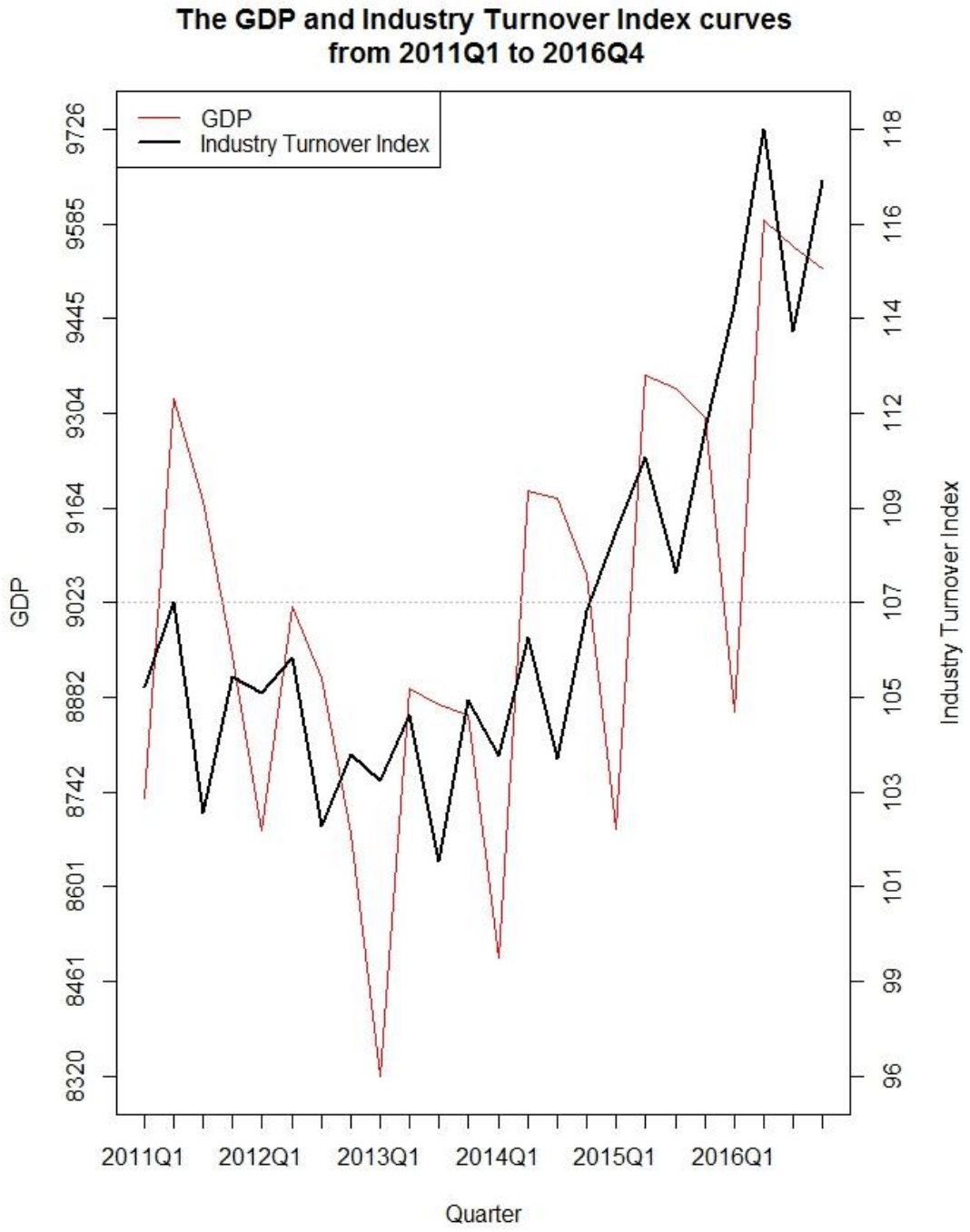
Period	Official values of	PCA method	No traffic data	Traffic data as secondary	Absolute values of	Absolute values of

	GDP (in million EUR)		estimates	regressor estimates	relative errors of the 1st est. (in %)	relative errors of the 2nd est. (in %)
2016Q2	9725,868	*75%*	9596,824	9640,344	1.33	0.88
		80%	9576,406	9708,404	1.54	0.18
		85%	9525,735	9614,310	2.06	1.15
		90%	9594,405	9630,760	1.35	0.98
		po10	9523,429	9548,396	2.08	1.83
		zadnja5	9678,580	9713,673	0.49	0.13
2016Q3	9682,643	*75%*	9613,220	9617,677	0.72	0.67
		80%	9630,709	9640,059	0.54	0.44
		85%	9632,367	9628,078	0.52	0.56
		90%	9693,590	9679,113	0.11	0.04
		po10	9400,231	9545,563	2.92	1.42
		zadnja5	9631,749	9640,023	0.53	0.44
2016Q4	9647,458	*75%*	9520,605	9546,125	1.32	1.05
		80%	9702,478	9635,077	0.57	0.13
		85%	9551,981	9480,159	0.99	1.73
		90%	9653,184	9541,028	0.06	1.10
		po10	9366,567	9439,871	2.91	2.15
		zadnja5	9544,756	9590,680	1.07	0.59

Source 3: Own

While these are really promising results, we are aware that an assessment on three temporal points is far from optimal. This is why we searched for a longer time series where traffic data could be used. With the assumption that industry turnover data represent a big part in the evaluation of GDP and therefore the two might correlate well, we tried using the traffic data as primary or secondary regressors for Industry Turnover Index nowcasts. Encouraged by the results above we expected that traffic data could give good assessments.

Picture 3: GDP and Industry Turnover Index time series



As before, we tested different linear regression with PCA models with different parameters. Furthermore we also used different sets of data for regression variables. We decided to test industry data as fitted with industry data alone, industry and traffic data as primary regressors, industry data as primary and traffic data as secondary regressors and traffic data alone as primary regressors. The learning phase was conducted on the data from 2011 to 2014 (64 temporal points) and the testing phase was conducted on the remaining 24 temporal points. With a larger pool of results we could also improve our criteria for an optimal model. We decided to choose the best model according to the *Root Mean Squared Forecast Errors* (RMSFE), which takes into account the accuracy of the

forecast in every testing temporal point and also their variability. Therefore the best model is such that has the best combination of closest average to the true values and the minimal deviation from that value in every temporal point in its monthly forecasts in a pre-agreed period.

The actual procedure was to test each parameter at the end of every year, and use the acquired optimal setup when nowcasting monthly values in the following year. An example of this will be given with the next table.

Judging from the RMSFEs of the models (in Table 4), the results are the same as in the above testing. In terms of data combinations, the best appear to use both the industry and traffic data. Meanwhile the best PCA parameters seem to be either *70%* or *80%*. The *zadnja5* model is also close to being the optimal model.

The following table shows the RMSFE of tested models:

Table 4: Root Mean Squared Errors for some linear regression - PCA models

	PCA: *70%*; Primary: Industry, Traffic	**PCA: *70%*; Primary: Industry; Secondary: Traffic**	**PCA: *80%*; Primary: Industry, Traffic**	**PCA: *80%*; Primary: Industry; Secondary: Traffic**	**PCA: *90%*; Primary: Industry, Traffic**	**PCA: *90%*; Primary: Industry; Secondary: Traffic**	**PCA: *zadnja5*; Primary: Industry, Traffic**	**PCA: *zadnja5*; Primary: Industry; Secondary: Traffic**
2015	2.90	2.50	2.25	2.84	4.45	5.97	2.82	2.47
2016	2.25	2.74	2.03	3.28	3.58	3.11	4.14	2.94

Source 4: Own

Based on the results of testing, in both years we would pick the model with the PCA parameter *80%* and both sets of data as primary regressors as our optimal model. The actual assessment of the Industry Turnover Index would be produced in the following way:

1. At the end of 2015 we would test different models and according to the results choose the optimal model (PCA: *80%*, both sets of data as primary regressors).
2. We would use this model for every monthly nowcast of 2016 and declare the assessments as our flash estimates.
3. At the end of 2016 we would again test all the models and choose the new optimal model (in this case the same one) to be used throughout 2017.

Using this strategy our estimates for the indices compared to their real values in the months of 2016 are:

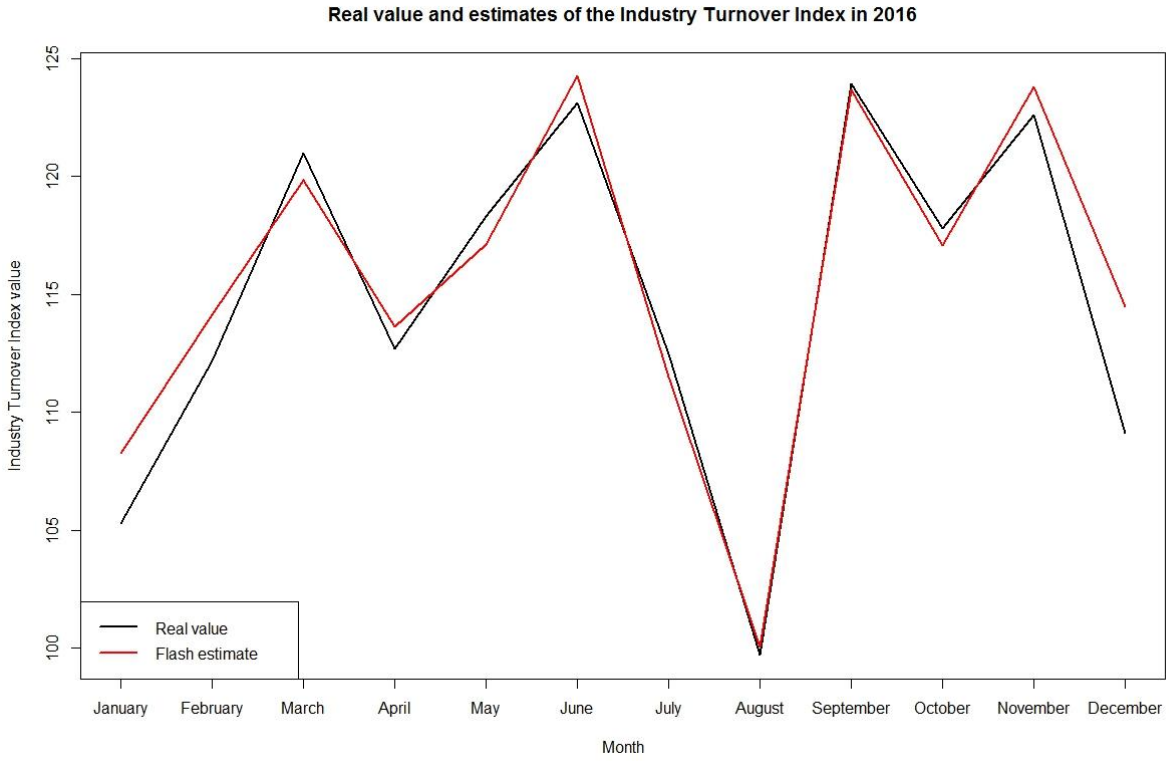
Table 5: Real values and estimates of the Industry Turnover Index in 2016

Month	Real value of the Industry Turnover Index	Flash estimate of the Industry Turnover Index	Absolute Errors	Relative Errors (in %)
January	105.3	108.27	2.97	-2.82
February	112.2	114.12	1.92	-1.71
March	121.0	119.82	1.18	0.98
April	112.7	113.63	0.93	-0.83
May	118.3	117.11	1.19	1.01
June	123.1	124.25	1.15	-0.93
July	112.5	111.51	0.99	0.88
August	99.7	100.05	0.35	-0.35
September	123.9	123.68	0.22	0.18
October	117.8	117.08	0.72	0.61
November	122.6	123.79	1.19	-0.97
December	109.1	114.47	5.37	-4.92

Source 5: Own

We can see that in most cases the relative errors are in the 1% limit, and the largest relative error is less than 5% off of the true value. This can be seen even better on the graph of comparisons of the two series below.

Picture 4: Comparison of the real values and estimates of the Industry Turnover Index in 2016



3 Machine Learning Approach and Forecast Combinations at Statistics Finland

In our empirical exercise, we use more than one hundred models in order to compute early estimates of the indicators of interest. Moreover, the theory behind a large part of these methods is fairly complicated. In the light of these considerations, we approach these methodological sections in a non-technical manner, giving the basic information and intuitions behind the adopted models. We discuss the generalities of the model classes we have explored, and then provide a list of the models which we incorporate in the forecast combinations, an approach which provides the best performance and is therefore the preferred methodological choice in nowcasting with the type of data we employ. Some practical considerations related to the overall methodological framework adopted in this study. All the models we use are easily implementable, in terms of software used and coding efforts. The machine learning techniques we adopt are all available within the 'caret' package, implemented in the free software environment 'R'. One benefit of 'caret' is that it allows an easy estimation of these models, while conducting an automatic model selection, without the need of an extensive input of the statistician. We run trials from the following categories of models, select the best performing ones (the ones that produce statistically unbiased estimates of the target indicator (mean error close to 0), and update the list of best models at a monthly interval.

3.1 Methodology

3.1.1 Factor models

The main idea underlying factor models is that a small number of constructed variables, factors, can summarize most of the information contained in a large dataset. This approach, together with principal component analysis, has a long tradition in statistics and econometrics. Principal component analysis was introduced by Pearson (1901) and Hotelling (1933), and it has been adopted in a wide range of applications, in psychology, engineering and economics, among others. The estimation strategy is described in the Slovenian discussion.

3.1.2 Linear regularized regressions

While the factor model described in the previous subsection solves the *curse of dimensionality* by extracting a relatively small number of variables from our large dimensional dataset, resulting in a two-step procedure, shrinkage methodologies regularize the coefficients of the original regressors. We examine three regularized regressions approaches, namely the ridge regression, the lasso and the elastic-net. One similarity among these models is that the regressors are included linearly.

Ridge regressions are specified as

$$(\mathbf{y} - \mathbf{X}\beta)'(\mathbf{y} - \mathbf{X}\beta) + \lambda \sum_{j=1}^{K} \beta_j^2,$$

Lasso regressions are

$$(\mathbf{y}-\mathbf{X}\beta)'(\mathbf{y}-\mathbf{X}\beta)+\lambda\sum_{j=1}^{K}|\beta_j|.$$

The main difference between these two is that in Lasso the irrelevant parameters are estimated to be near zero, leading to model selection setting. Introduced in Zou and Hastie (2005), the elastic net combines ridge-regression and the lasso. It is based on the following minimization problem

$$(\mathbf{y}-\mathbf{X}\beta)'(\mathbf{y}-\mathbf{X}\beta)+\lambda_1\sum_{j=1}^{K}|\beta_j|+\lambda_2\sum_{j=1}^{K}\beta_j^2$$

One of the main benefits of the elastic-net is that it is better suited in a scenario where the regressors are strongly correlated, and it has been shown to work better when the number of regressors is larger than the number of observations.

3.1.3 Non-linear models

In our study, we have examined the nowcasting ability of a large number of machine learning methods, going from tree-based models to boosting and neural networks. We are not going to offer a thorough examination of these techniques; however, we go over the main intuitions and principles underlying the main families of machine learning methods that we have adopted. A much more detailed discussion of these models can be found in Hastie, Tibshirani, and Friedman (2009).

3.1.3.1 Boosting

Boosting is a form of forward stage-wise modelling, where our target variable of interest y_t can be expressed as an additive function

$$\hat{f}_M(X_t)=\sum_{m=1}^{M}b(X_t,\hat{\beta}_m),$$

for t = 1, . . . , T, where T is the number of observations we have. In (9), $b(X_t,\hat{\beta}_m)$ are called *learner* and are a, possibly non-linear, function of the regressors. M represents the total number of boosting iterations which governs how the final model fits the data. Notice that the boosting procedure is feasible in a high-dimensional setting because for each iteration m the parameters estimated in the previous iteration are left unchanged.

3.1.3.2 Tree-based methods

Tree-based techniques partition the space of explanatory variables in order to fit a simple model for each partition. To make the idea clearer, let's proceed with a very simple example. Assume that we have a variable Y which is a simple linear function of an individual regressor X plus a normally distributed error. For this kind of scenario, linear regression models would work just fine but this kind of trivial application can be useful to grasp the intuition behind regression trees. In a basic

regression tree, we split the X space in different regions and fit a constant model for each region. Formally, assume that we have P partition of the X space, then we have

$$\hat{f}(X) = \sum_{p=1}^{P} c_p I\{X \in R_p\},$$

where $I(X,R_p)$ is an indicator function which is equal to 1 if the X belongs to partition R_p. It can be shown that under squared loss function the optimal estimate of c_p is simply the average of Y conditional on X belonging to R_p. In our example, we simulate 100 observations of the aforementioned process and fit a simple regression tree model. The resulting scatterplot and the graphical representation of the tree are reported below.

Picture 4: Regression tree technique

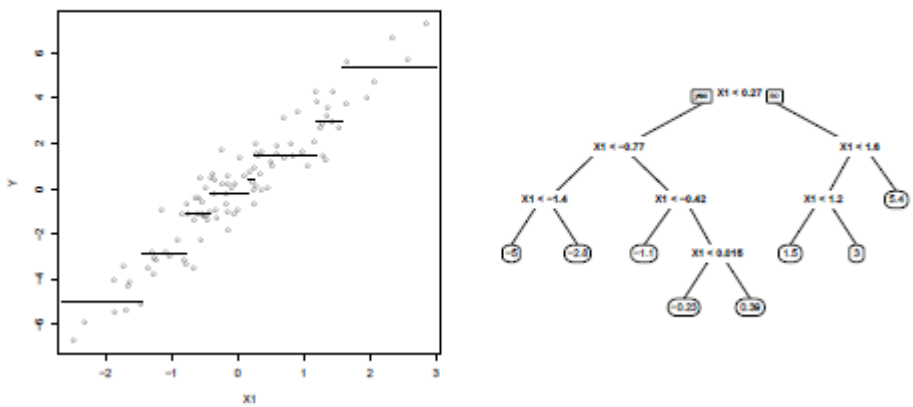


Figure on the left gives a fairly clear representation of the regression tree technique. It separates our X values into 7 regions (corresponding to seven splits). For each region, we calculate the average Y so that when we get a new value of X the corresponding predictions Y will simply be the average of Y corresponding to region p to which the new X belongs. For example, if our new X is between -0.42 and 0.015, then the predicted Y will be -0.23. Naturally, regression trees can deal with numerous regressors, which will then impact the optimal splits and tree size. Notice that we need to estimate the optimal splits of the regressors and the depth of the tree. While the first aspect is estimated, how large we should grow the tree is left as a tuning parameter. A typical strategy is to grow a very large tree and then prune the tree afterward, to reduce its complexity.

3.1.3.3 Neural Networks

Despite their somewhat mysterious name, neural networks are just a class of nonlinear statistical models, which can be split in two stages. In the first step, a set of derived features Z_m are created from linear combinations of inputs (our Xs). Subsequently, our target variable Y is modelled as a function of Z_m. Formally, we would have

$$Z_m = \beta_0 + \beta_m X, \text{ for } m = 1, \ldots, M$$
$$Y = f(\alpha_0 + \alpha Z)$$

The non-linear aspect of neural networks is introduced with $f(Z)$, where the function $f(\cdot)$ can take many forms. One of the most typical choices for these so-called activation functions is the *sigmoid*

$$\sigma(x) = 1/(1 + e^{-x}).$$

Given the neural network structure, we need to estimate a set of parameters often called *weights*, which we denote as θ. In the setting where we have K inputs and M derived features, the complete set of weights is given by:

$$\{\beta_0, \beta_m; m = 1, 2, \ldots, N\} \; M(K+1) \text{ weights,}$$
$$\{\alpha_0, \alpha\} \; (M+1) \text{ weights.}$$

In the regression setting, we use the sum of squared-errors.

$$R(\theta) = \sum_{t=1}^{T} (Y_t - f(X_t))^2$$

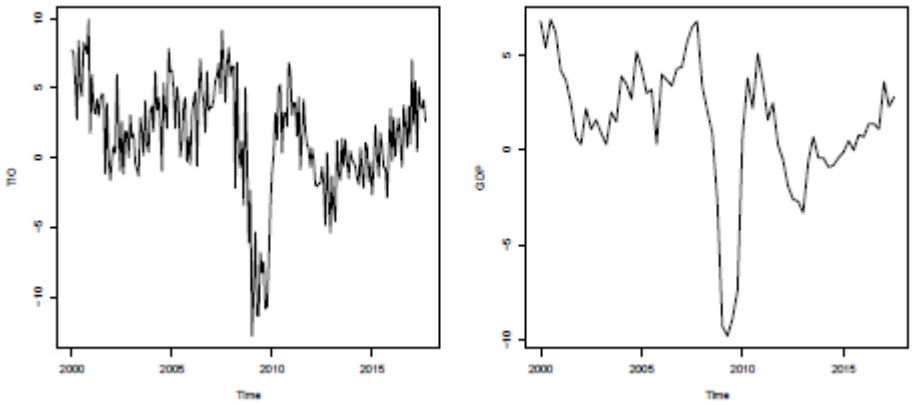
The typical procedure adopted in order to minimize the error function is by gradient descent, which is typically called *back propagation*.

3.2 Nowcasting application description

The main regressors in our nowcasting application are firm-level sales extracted from the sales inquiry, a monthly survey conducted by Statistics Finland for the purposes of obtaining turnovers from the most important firms in the economy. This dataset covers around 2,000 enterprises and encompasses different industries (services, trade, construction, manufacturing), representing about 70% of total turnovers. The data are available soon after the end of the month of interest and a considerable share of the final data is accumulated around 15 to 20 days after the end of the reference month. Formally, Statistics Finland imposes a deadline to the firms, which are supposed to send their data by the end of the 15th day of the month. We compute the nowcast on the 16th day. However, this deadline is not always met, thus our set of firms' sales does not cover the entire sample. The data accumulation is realistically simulated by using the time stamp of the reported sales, which allows us to track what data were available by each date of a month. We require that firms have long time series (starting in 2006), and that they have reported sales figures by the date we extract their information from the database. By choice, we collect data of the firms that have reported the sales by 16 days after the end of the reference month, leading us to have 800 firms on average in the regressors' set. We compute the sales growth rates for all the months from 2006 until the nowcasted month of interest. If the firm has reported sales by the t + 16 at the nowcasted month, but has missing values during the time span (i.e. the firm did not reply at some earlier date, or the firm was not included in the turnover inquiry at that time), we try to obtain the missing growth rates from VAT data, which should include all the firms in the economy. Notice that our resulting data do not contain missing values. The target variables in our exercise are the Trend Indicator of Output (TIO) and quarterly GDP, both measured in real-term year-on-year growth rates.

The TIO is a monthly series that describes the development of the volume of produced output in the economy. It is constructed by using early estimates of turnover indices (not publicly available), which are appropriately weighted to form the monthly aggregate index. The TIO is published monthly at t + 45, and its value for the third month of a quarter is used to compute the flash estimate of GDP, which is also published as an early version at t + 45, and updated at t + 60. The t + 60 version is considered as the first official and reliable estimate of GDP. Thus, given the information we have provided, the TIO in fact represents a GDP nowcast in its own right. We stress the importance of using the realistic vintages, as the data are constantly "improved" by many internal processes, and by the accumulation of new data. The usage of revised data can arguably lead to too optimistic views on the nowcasting performance of our models and consequently invalidate the results, especially in the eyes of the statistical office staff members who we wish to convince about the usefulness of the proposed methodology. The methods need to be working with the realistic data that the official statisticians have to deal with. Below we report the plots of the TIO and GDP year-on-year growth rates.

Picture 5: TIO and GDP year-on-year growth rates



One aspect that it is important to underline is how closely related the TIO and GDP growth are. If we aggregate TIO growth to the quarterly level we obtain a series that closely tracks GDP growth (the resulting correlation coefficient is 0.99). This demonstrates that providing a good estimate of TIO leads to a greater nowcasting accuracy of GDP.

3.3 Traffic loop data

One of the main objectives of the ESSnet Big Data Early Estimates working group is to investigate the nowcasting potential of big data sources. The firm-level data which constitute the main data source of our exercise provide a good nowcasting performance. However, while high-dimensional, the firm-level turnovers we use are not a big data source in a traditional sense. In particular, they are not collected in real-time, i.e. firms send their data after the end of the reference month. Moreover, our turnover dataset is structured and fairly easy to handle, which is not typical of big data. In addition to the nowcasting exercise involving firm-level data, we consider the predictive performance of traffic volumes records obtained by the Finnish Transport Agency website. This dataset contains the number of vehicles passing through a number of measurement points (about 500) around Finland,

observed through an automatic traffic monitoring system. The data are available at hourly frequency, and it distinguishes between different types of vehicles. This dataset contains numerous missing values, due to the fact that some measurement points do not have observation for certain days or months, and it is not structured. For our nowcasting analysis, we collect data for trucks' traffic volumes from January 2010 (the first dataset available), in particular their year-on-year growth rate. Trucks' traffic presents an interesting intuitive link with aggregate economic activity, especially with trade. We expect that in periods of economic growth, where trade volumes increase across the country, we should observe a higher number of trucks' passages, in order to move goods. Of course, this does not cover the transfer of services and other types of economic activities, but it should still present some positive correlation with economic activity growth. Given that the data are available only from January 2010, we have decided to start the computation of pseudo-real-time nowcasts of TIO growth from January 2014, to give us four years of estimation sample. Similarly as in the firm-level data case, we adopt the predicted TIO growth rates to compute the year-on-year growth of GDP.

The traffic data are aggregated at the monthly level and we assume that our estimation of TIO is conducted around 16 days after the end of the reference month (as in the main exercise). This allows us to use the Statistics Finland's estimates of TIO for the $t-1$ month, where t represents the period we want to nowcast. However, in principle the traffic data we utilize allow for nowcasts during the month of interest, given their daily frequency. It is important to point out that, unlike the firm-level data we utilize, our set of traffic volumes contains missing values. In order to impute the missing observations, we rely on the regularized principal component technique illustrated in Josse and Husson (2016). The actual nowcasts are computed using statistical models and machine learning techniques similar to the ones described in Section 2. The final nowcasts are obtained by making a simple unweighted average of the individual predictions. After trimming the modes producing large historical mean errors, this procedure yields us around 20 different models.

3.4 Quality Assessment of Nowcasting Performance

Given that our aim is to form a strategy by which we can forecast the near past and the present evolution of the target indicators, we have to evaluate the quality in terms of timeliness and accuracy in capturing the target indicators' next values. The standard measures of accuracy in these types of exercises are the Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE), but we also consider Mean Error (ME), and the Maximum Error (MaxE). ME is an important measure, because it indicates the unbiasedness of our estimate vs the target indicator. MaxE can be important to the NSIs, because very large errors in a widely followed GDP series can be a cause for negative publicity. MAE and RMSE are the standard measures of deviations from the target (forecast errors), and RMSE gives relatively more weight on very large errors.

Results are compared to the next version of the published target indicators, precisely because we want to produce estimates of the next value of the target indicator, which are routinely published by the NSIs. In this way, the nowcasts represent a natural accumulation of data. We know that the first estimates are widely followed by the forecasting institutions as well as other sophisticated users, who understand that the GDP series are revised once more data are accumulated. To them, an important aspect of the estimate is that it is not possible to statistically distinguish the nowcast from

the target. This is why we refer to the measure of unbiasedness (ME) and provide a regression of target Y onto the nowcasted value of Y, to show that the residuals are white noise.

A very useful point of assessing the accuracy dimension in the quality of nowcast is to compare the nowcasting accuracy to the revisions of the statistical office. In other words, to the standard that is acceptable by the NSI for a regular dissemination. We do this by looking at the revisions in the published series (the final available value of the series growth rates vs the initial estimates by the statistical office). We infer that our nowcasting strategy is capable of producing a much earlier estimate of the GDP (by 1.5 months) without sacrificing too much in terms of accuracy. (The accuracy is similar as in the official first releases).

Picture 6: Residuals are white noise from regressing the official first estimate vs the nowcasted TIO

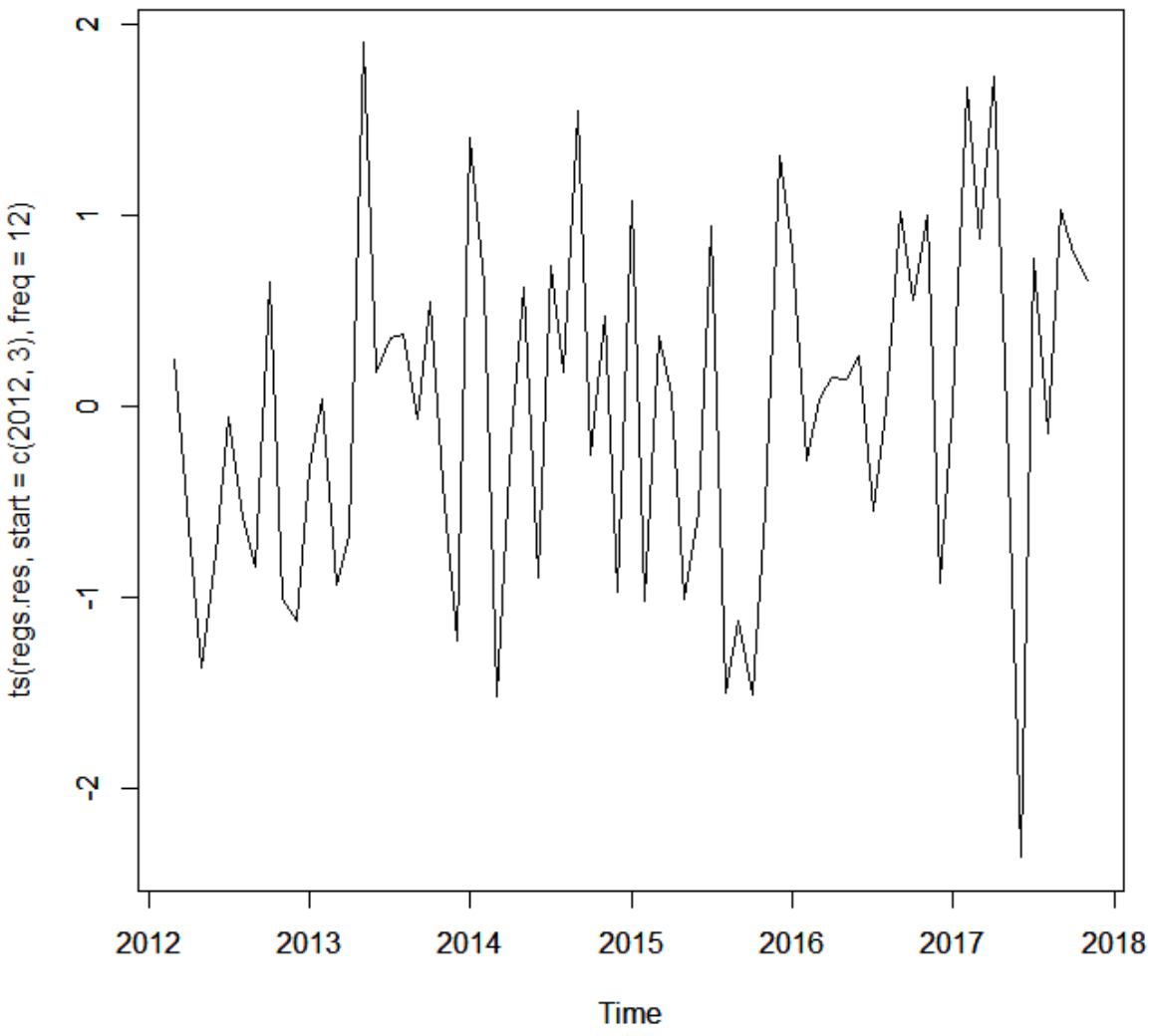


Table 7: The accuracy measures of TIO for estimates at t+16. Results from five modelling strategies are presented here, the results from the model which produces the lowest mean error (Lowest ME), the lowest RMSE, the lowest MAE, lowest MaxE, and the simple forecast combination

	Lowest ME	Lowest RMSE	Lowest MAE	Lowest MaxE	Combination
ME	0.00	-0.05	-0.05	0.00	-0.00
MAE	0.83	0.81	0.81	0.83	0.76
RMSE	1.02	0.98	0.98	1.02	0.93
MaxE	2.36	2.64	2.64	2.36	2.55

This table demonstrates that forecast combination seems to be the most desired approach (lowest MAE, RMSE and low Max E). Notice that Statistics Finland makes roughly 0.9 percentage point revisions to the TIO.

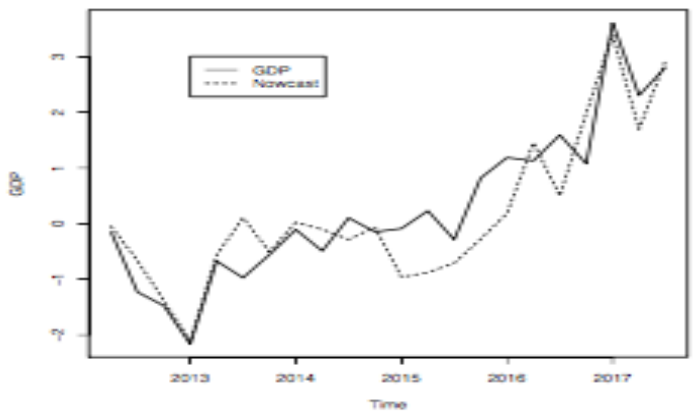
Table 8: The accuracy measures of GDP quarterly for estimates produced during the reference quarter and after (t+16).

	Nowcast second month	Nowcast third month	Nowcasts 16 days after
ME	-0.14	-0.03	-0.12
MAE	1.07	0.66	0.49
RMSE	1.28	0.87	0.63
MaxE	2.58	2.17	1.11

Notice, that Statistics Finland makes roughly 0.5 percentage point revisions to the GDP quarterly series.

3.5 The Example of t+16 Estimate of the Quarterly GDP (Y-o-Y) vs. t+60 Official Release

Picture 7: Early Estimate of GDP vs the Official Release



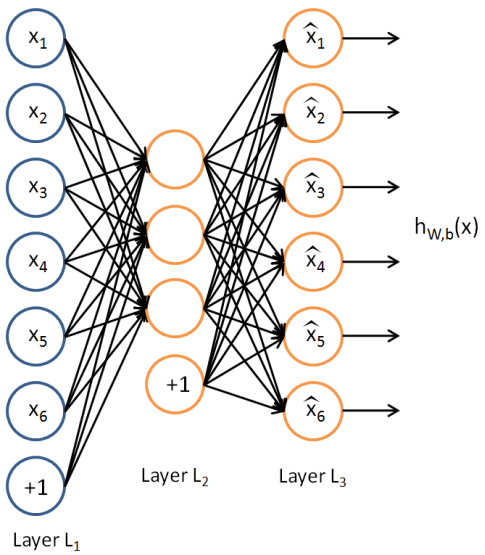
We manage to create early estimates with similar accuracy as the official release 1.5 months earlier. The above figure shows results which correspond to 0.49 percentage point MAE in relation to the

t+60 publication by the statistical office. The early estimate is obtained by aggregating the monthly TIO nowcasts to the quarterly level. The process can be automated almost fully, and is applicable in many countries interested in producing Early Estimates in this way, using disaggregated micro-level sources.

4 Using Monte Carlo Markov Chain (MCMC) algorithm at Statistics Netherlands

In the Netherlands, a different approach was developed. Due to the large number of traffic loops (20,000 on the Dutch highways), another data cleaning strategy had to be developed. To enhance the speed of the data cleaning process, a Monte Carlo Markov Chain (MCMC) algorithm was used to clean the data, remove noise and solve the problem of missing data (see [11]) for a description of the complete process). Principal component analysis cannot be used due to several reasons. The reason is that the dataset has more columns than rows when looking at daily data. Since Statistics Netherlands had data since 2011, this means that less than 2,500 days of data are available, whereas 20,000 measurements are available per day. This means that the PCA has to be performed on a 2,500 x 20,000 matrix. Furthermore, although many minutes of data are reconstructed using the MCMC algorithm, for many days road sensors are missing data and with PCA it is hard to deal with missing data.

Due to the reasons mentioned above, Statistics Netherlands is applying a more novel approach as an alternative for PCA. In this approach, an autoencoder is used for the dimension reduction (see http://ufldl.stanford.edu/tutorial/unsupervised/Autoencoders/ for a tutorial). An autoencoder is a neural network that learns to reconstruct a certain input on the output. The hidden layer consists of a number of units which is much smaller than the amount of input units (see figure).



By adding extra hidden layers, and by applying training techniques (drop out, regularization and adding Gaussian noise), the neural network is able to deal with missing and erroneous values, which makes it ideal for data cleaning and dimension reduction at the same time. By training subsets of

road sensors in different neural networks, the dimensionality problem as described above is solved. The reduced dataset can then be used for regression purposes.

Since this is still work in process, we are not yet able to present the results in this document.

5 Conclusions

We find that using incomplete early micro-level sources and a real-time big data source such as the traffic loop data can be used to produce early estimates of economic indicators.

What we gathered during our processing of data and the errors is that quite a lot of errors arise from specific characteristics of given datasets (metadata errors, weighting circumstances, editing) and cannot be generalized to the whole or at least to a big part of the Big Data field. The solutions to most of these individual problems need to be specifically made for them. Lucky for use the source of traffic sensor is fairly consistent and the solutions we devised in this pilot can be applied in the future as well without (much) changes. Furthermore some of these problems are sensor-specific and while each problem needs to be addressed individually, they are part of a common set of problems that arise when working with other smart sensors as well. As such a lot of solutions on other smart sensor data can probably be adapted from todays (a good example are missing data due to malfunctions).

In the case of traffic sensors, measurement errors mainly arise from sensors malfunctions and/or switching off. This leads to partly or entirely non-existent data for a period. Missing data (such as missing periods or empty variables) can be a clue as to whether there was a malfunction or not. Other ways of detecting periods of malfunction are finding suspicious uncharacteristic values in the time series of data or variables wholly comprised of zeroes. Unfortunately, this data coming from a second-hand source, we cannot exactly know when a sensor malfunctioned, especially if the malfunction occurred during one interval. It's a fact that we cannot know if a sensor is consistently wrong in their counting unless we perform a manual count on the same road as the sensor.

In case of dealing with large number of traffic loops, a more novel approach as an alternative for PCA should be considered for dimension reduction. The neural network is able to deal with missing and erroneous values, which makes it ideal for data cleaning and dimension reduction at the same time. By training subsets of road sensors in different neural networks, the dimensionality problem as described above is solved. The reduced dataset can then be used for regression purposes.

In some years enough quarters will pass to efficiently use RMSFE criteria for optimal model selecting on GDP. At that moment we will be able to accurately assess whether is it better to flash estimate directly GDP or if we should focus on rapid estimations of its less-timely components and then use them as regular data in the normal estimation process for GDP. In addition it has to be investigated what precision is required in order to produce reliable earlier estimates, explore the possibilities to nowcast only some components of quarterly GDP (for which all data are not available on time), include other big data (and other) sources and also test some additional methods for nowcasting.

Given that our aim is to form a strategy by which we can forecast the near past and the present evolution of the target indicators, we have to evaluate the quality in terms of timeliness and accuracy in capturing the target indicators' next values. The standard measures of accuracy in these types of exercises are the Mean Absolute Error (MAE), and Root Mean Squared Error (RMSE), but we also consider Mean Error (ME), and the Maximum Error (MaxE). ME is an important measure, because it indicates the unbiasedness of our estimate vs the target indicator. MaxE can be important to the NSIs, because very large errors in a widely followed GDP series can be a cause for negative publicity. MAE and RMSE are the standard measures of deviations from the target (forecast errors),

and RMSE gives relatively more weight on very large errors. A very useful point of assessing the accuracy dimension in the quality of nowcast is to compare the nowcasting accuracy to the revisions of the statistical office. In other words, to the standard that is acceptable by the NSI for a regular dissemination. We do this by looking at the revisions in the published series (the final available value of the series growth rates vs the initial estimates by the statistical office).

It is expected that turnover indicators will become timelier and (what is remarkable) more accurate, which in turn makes it possible to produce the aggregate output indicators faster and more accurately by the existing national accounts procedures. We also provide a new methodology by which an early estimate of economic indicators can be produced reliably (quantitatively with similar accuracy as the official first estimates) by employing micro-level turnovers and traffic loop data together with various machine learning methods designed to handle such large dimensional data. These results have several implications. Firstly, the statistical office can easily meet the Eurostat requirements for timelier delivery of the flash GDP. Secondly, one can potentially construct real time and possibly high frequency (e.g. daily) estimates of the economic output. Thirdly, the traffic loop data should be inspected further for instance by exploring clusters of the measurement points around designated areas (borders, manufacturing clusters, mining fields, etc.). Further, the explanatory power of traffic data should be examined in relation to the turnover indicators or other specific sectors.

6 Literature

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